

Akagera Group Assignment

Topic: DATA CARD

Nigerian Agriculture & Climate Small Language Model Dataset

Data Card Dimension	Project Specification
Dataset / Project Name	Nigerian Agriculture and Climate AI Dataset (NAC-AI Dataset)
Associated Model	Small Language Model (SLM) for Agriculture and Climate Advisory
Primary Domain	Agriculture, climate change, climate-smart agriculture, weather, soil, crops, livestock, fisheries and rural livelihoods
Geographical Focus	Nigeria, with emphasis on rural and smallholder farming communities
Primary Beneficiaries	Smallholder farmers, agricultural extension officers, farmer cooperatives, agricultural educators, NGOs and rural development organizations
Primary Purpose	To train and evaluate an AI model capable of providing accurate, contextualized, culturally appropriate and actionable agricultural and climate information for Nigerian farming communities.
Intended Use	Agricultural information retrieval, farmer education, climate-risk awareness, crop management guidance, pest/disease information, climate-smart agriculture advisory and extension support.
Non-Intended Use	The model should not replace qualified agronomists, veterinarians, meteorologists, extension professionals or emergency authorities, particularly for high-risk decisions.

Classification Problem

The project can be conceptualized as a domain-specific language modelling and classification problem.

Primary classification tasks

Classification Task	Possible Classes/Labels
Agricultural domain	Crop production, Livestock , Fisheries , Agroforestry , Soil / Irrigation , Post-harvest
Climate information	Drought, Flood, Heat, Excessive rainfall, Dry spell, Normal condition
Crop growth stage	Land preparation, Planting, Vegetative, Flowering, Fruiting, Harvest, Post-harvest

Farmer query type	Information seeking, Diagnosis, Recommendation, Prediction, Problem reporting
Pest/disease category	Pest, Fungal disease, Bacterial disease, Viral disease, Nutrient deficiency, Abiotic stress
Climate-smart practice	Water management, Soil conservation, Crop diversification, Improved varieties, Agroforestry, Weather-informed farming
Risk level	Low, Moderate, High, Critical
Advisory action	Monitor, Prevent, Treat, Adapt, Seek expert assistance
Knowledge source	Scientific, Government, Indigenous-local, Extension, Farmer-generated
Response quality	Correct, Partially correct, Incorrect, Unsafe

Core ML formulation

The model should ultimately learn the relationship: Farmer context, agricultural problem, environmental/climate context, relevant knowledge, appropriate recommendation. Rather than simply learning to produce plausible text, the dataset should therefore contain context-rich examples that teach the model why a particular recommendation is appropriate.

Labelling Method

A **multi-level annotation framework** is recommended.

Level 1: Metadata labelling

Each record should be captured where available:

- State
- Agroecological zone
- Local Government Area
- Crop/livestock system
- Farming system
- Season
- Climate/environmental condition
- Farmer type
- Source

- Date
- Language

Level 2: Intent labelling

Each farmer query can be assigned an intent such as:

QUERY, INTENT, DOMAIN, CONTEXT and RESPONSE

For example:

"My maize leaves are turning yellow after several days of heavy rain. What should I do?"

Possible annotation:

Intent: Problem diagnosis

Domain: Maize production

Stress: Excessive rainfall

Symptom: Leaf yellowing

Potential causes: Waterlogging/nutrient-related stress/disease

Risk: Moderate

Required response: Differential diagnosis, recommended action, and expert referral where necessary.

Level 3: Response-quality labelling

Researchers and annotators will evaluate generated or collected responses using criteria such as:

1. Factual accuracy
2. Agronomic validity
3. Climate relevance
4. Local applicability
5. Actionability
6. Language clarity
7. Safety
8. Cultural appropriateness

A simple scale could be:

0 = Incorrect/unsafe

1 = Partially correct

2 = Correct but incomplete

3 = Correct, complete and contextually appropriate

Recommended annotation approach

Use dual independent annotation and adjudication:

Annotator A, Annotator B, Agreement assessment, Expert adjudication, and Final label

Inter-annotator agreement can be assessed using Cohen's κ for categorical labels or Krippendorff's α where multiple annotators/label types are involved.

Data Creation

The dataset should not rely on a single source. A hybrid data-generation strategy is preferable.

A. Real-world data

The Researcher will collect naturally occurring agricultural questions and responses from:

- Farmer interviews
- Focus-group discussions
- Agricultural extension interactions
- Farmer cooperatives
- Agricultural call centres
- Participatory rural appraisal
- Community workshops
- Existing agricultural information systems

B. Expert-generated data

Agronomists, soil scientists, climate scientists, agricultural extension officers and other domain experts can generate:

- Question–answer pairs
- Crop-management scenarios
- Climate-risk scenarios
- Pest/disease cases
- Seasonal farming scenarios
- Frequently asked questions

- Corrective examples of common misconceptions

C. Synthetic data

Researchers could use LLM to generate candidate examples, but these should not automatically become ground truth.

Recommended workflow:

Expert knowledge, Seed examples, Controlled synthetic generation, Expert review, Validation, and Dataset

This is particularly important because synthetic data can amplify the hallucinations and biases of the model that generated it.

D. Data augmentation

Possible augmentation methods include:

- Paraphrasing
- Question reformulation
- Translation
- Back-translation
- Farmer-language variations
- Spelling/noisy-text variations
- Voice-to-text transcription
- Scenario generation

However, augmented records should retain a link to their original source record to prevent accidental inflation of evidence.

4. Likely Sources of Data Collection

A strong Nigerian dataset should deliberately combine scientific, institutional, community and indigenous knowledge.

Source Category	Examples of Data
Government agricultural agencies	Agricultural policies, extension manuals, crop production guides
Nigerian Meteorological Agency (NiMet)	Weather and climate information
Federal Ministry of Agriculture and Food Security	Agricultural guidelines, programmes and technical documents
National Bureau of Statistics	Agricultural and socioeconomic statistics

Agricultural Research Institutes	Crop, soil, pest, livestock and production knowledge
Universities	Theses, dissertations, research publications and extension materials
International organizations	FAO, CGIAR and related agricultural/climate resources
Climate organizations	IPCC and climate-data/research resources
Extension services	Farmer advisory materials and field knowledge
Farmer cooperatives	Local practices, challenges, questions and experiences
Farmer interviews	First-hand knowledge and decision-making practices
Focus groups	Collective community knowledge and perceptions
Indigenous knowledge holders	Locally developed climate indicators and agricultural practices
Open-access scientific literature	Peer-reviewed agricultural and climate research
Satellite/environmental data	Land cover, vegetation, rainfall, drought and environmental indicators
Public weather datasets/APIs	Historical and near-real-time weather information
Agricultural NGOs	Training manuals, field reports and community-level agricultural information

Data should only be incorporated where licensing, copyright, terms of use, privacy and research-ethics requirements permit it.

5. Ethics and Responsible Data Governance

Ethics should be treated as a core component of dataset design, rather than something added after collection.

Informed consent

Ethics should be treated as a core component of dataset design, rather than something added after collection.

Privacy

We avoided storing unnecessary:

- Names
- Telephone numbers
- Exact household addresses
- Bank/financial information
- Identification numbers

- Personally identifiable information

Where personal information is inadvertently captured, it should be redacted or pseudonymized before model training.

Community ownership and knowledge rights

Particular care will be taken for the indigenous and community knowledge. The project should distinguish between: Public knowledge, Open community knowledge, Restricted community knowledge, Sensitive/traditional knowledge. Community knowledge should not automatically be treated as freely exploitable training data merely because it was obtained during an interview. Hence, the researcher will be cautious in handling these data.

Potentially harmful advice

The model was designed to recognize situations requiring professional referral, including:

- Severe crop disease outbreaks
- Toxic chemical exposure
- Unsafe pesticide practices
- Livestock disease emergencies
- Extreme weather events
- Flood emergencies
- Financially consequential decisions

6. Bias and Representation

This is one of the most important components of the dataset.

A Nigerian agricultural AI system can become systematically biased if its training data disproportionately represent a few regions, crops, languages or farming systems.

Potential sources of bias

Bias	Potential Problem
Geographic bias	Data concentrated in Southern or Northern Nigeria
Agroecological bias	Some ecological zones poorly represented
Crop bias	Maize, rice and cassava dominate while minor crops are neglected
Language bias	English dominates over Nigerian/local languages

Gender bias	Male farmers overrepresented
Age bias	Younger digitally connected farmers dominate
Digital-access bias	Offline/rural farmers are underrepresented
Knowledge-source bias	Scientific knowledge dominates indigenous knowledge
Institutional bias	Government/academic perspectives dominate farmer experience
Socioeconomic bias	Commercial farmers receive more representation than subsistence farmers
Climate-event bias	Dramatic events such as floods receive more attention than gradual climate change
Label bias	Experts may interpret farmer knowledge differently from community members

Representation strategy

The dataset should deliberately construct a **representation matrix** covering: Six geopolitical zones, agroecological zones, major farming systems, gender, age groups, crops, languages, and rural contexts. The actual sampling proportions should be determined from the intended deployment population rather than assuming that every category must have equal representation.

Representation audit

Before training, the Researchers will calculate the distributions for:

- State
- Geopolitical zone
- Agroecological zone
- Crop
- Gender
- Age
- Language
- Farming system
- Data source

- Climate hazard
- Query type

A data coverage dashboard should identify underrepresented groups before model training.

7. Values and Community Alignment

The dataset should reflect the values of the communities it is intended to serve.

Core alignment principles

1. Farmer-centredness

The model was built to respond to farmers' actual problems rather than merely reproduce academic agricultural terminology.

2. Local relevance

Recommendations accounted for Nigerian farming conditions, resources, climate and socioeconomic realities.

3. Cultural respect

Local agricultural knowledge was documented respectfully and not automatically dismissed when it differs from conventional scientific knowledge.

4. Scientific integrity

Traditional knowledge and scientific knowledge should be distinguished where necessary rather than falsely presenting one as the other.

5. Sustainability

Recommendations encourage practices that protect soil, water, biodiversity and long-term agricultural productivity.

6. Inclusiveness

Women farmers, youth, smallholders, persons with limited literacy and marginalized rural communities should be represented.

7. Transparency

Where feasible, users should be able to distinguish between scientific evidence, community knowledge, model inference and uncertainty.

8. Human oversight

AI should function as an advisory and educational tool, not an unquestionable authority.

8. Recommended Data Schema

Each training example could follow a structure such as:

Field	Example
record	NG_AGRI_000001
location	Enugu State
agroecological zone	Derived Savannah
crop	Maize
farming system	Smallholder
season	Rainy season
language	English/Igbo
query	Farmer's question
intent	Crop problem diagnosis
climate context	Excessive rainfall
knowledge source	Extension officer
evidence level	Expert validated
response	Recommended advisory
risk level	Moderate
expert validation	Approved
cultural relevance	High
safety status	Safe
consent status	Consented
license	Permitted
annotator agreement	0.87

Field	Example
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quality score	3/3
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9. Data Quality-Control Pipeline

A robust pipeline would be: (i) Data acquisition (ii) Consent/licensing verification (iii) PII removal (iv) Data cleaning (v) Language identification (vi) Translation/transcription where necessary (vii) Domain (viii) classification (ix) Human annotation (x) Expert validation (xi) Bias & representation audit (xii) Safety assessment (xiii) Deduplication (xiv) Train/validation/test split (xv) Model training & evaluation. The test set should be protected from training-data contamination, particularly where synthetic generation is used.

10. Data Card Summary

Dimension	Commitment
Purpose	Build a Nigerian-focused agricultural and climate knowledge base for an SLM
Problem type	Domain classification, information retrieval, question answering and contextual agricultural advisory
Data types	Text, expert Q&A, farmer queries, extension materials, climate/agricultural data and potentially speech/transcriptions
Labelling	Multi-level human annotation with expert adjudication
Data creation	Real-world, expert-generated, and carefully validated synthetic data
Primary sources	Farmers, extension workers, research institutions, government agencies, universities, climate/agriculture organizations and open scientific resources
Output data type	Text-text, text-to-audio, audio-to-text
Ethics	Consent, privacy, licensing, community knowledge protection and responsible use
Bias control	Geographic, linguistic, gender, crop, socioeconomic and knowledge-source representation audits
Community alignment	Farmer-centred, culturally respectful, scientifically grounded, inclusive and sustainability-oriented
Human oversight	Required for high-risk or uncertain recommendations
Key risk	Model may produce plausible but agriculturally incorrect or locally inappropriate advice
Mitigation	Expert validation, evidence tagging, uncertainty communication, retrieval augmentation and continuous community evaluation
Success criterion	The model provides accurate, safe, locally relevant and understandable agricultural/climate information across diverse Nigerian farming communities.